

The Editor

Computer Networks
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Subject: Submission of Manuscript - "**FedSemGNN: A Scalable, Semantic-Aware Federated RL Framework for Efficient 6G Edge Orchestration**"

Dear Editor,

We are pleased to submit our manuscript entitled "**FedSemGNN: A Scalable, Semantic-Aware Federated RL Framework for Efficient 6G Edge Orchestration**" for consideration for publication in Computer Networks.

This paper presents a novel hierarchical federated reinforcement learning framework that integrates semantic-aware task representation, graph neural network topology encoding, and hierarchical Proximal Policy Optimization to achieve ultra-low latency, exceptional scalability, and privacy-preserving orchestration for 6G edge environments. The proposed FedSemGNN approach achieves orchestration latency of 0.309–0.328 ms with $0.967\times$ scaling factor demonstrating sub-linear computational complexity, exceptional energy efficiency (72.1 W), and perfect semantic fidelity (100% with zero migrations) across 100 to 10,000 simulated nodes ($100\times$ scale range), significantly outperforming state-of-the-art methods including ECO-SDIoT, GFL-LFF, and FRPVC (published in Computer Networks) by 55 to 4,444 times in latency reduction.

The main contributions of this work are as follows:

- Semantic-aware task representation using a custom continual learning encoder that produces 16-dimensional embeddings from service characteristics, enabling intent-aware orchestration beyond resource-based placement.
- Graph neural network topology encoding through a two-layer GCN with symmetric normalization that captures network structure, spatial relationships, and multi-hop dependencies for topology-informed decision-making.
- Hierarchical federated reinforcement learning with dual-level synchronization (local every 10 epochs, global every 50 epochs) that achieves $161\times$ communication reduction while maintaining convergence stability and privacy guarantees.
- Lightweight neural architecture with compact actor-critic design (16 hidden units) enabling sub-millisecond orchestration decisions suitable for real-time 6G applications.
- Comprehensive experimental validation through extensive simulations in EdgeSimPy with detailed performance analysis across reward, latency, power consumption, semantic fidelity, communication efficiency, and migration stability metrics, demonstrating exceptional scalability with sub-linear computational complexity.

To the best of our knowledge, this is the first study to jointly integrate semantic embeddings, GNN-based topology encoding, and hierarchical federated PPO within a unified orchestration framework for large-scale 6G edge systems. We believe this paper fits well within the scope of Computer Networks, as it addresses critical challenges in intelligent, scalable, and communication-efficient orchestration for next-generation edge computing infrastructures.

Thank you for considering our manuscript. We look forward to your favorable response.

Sincerely,
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